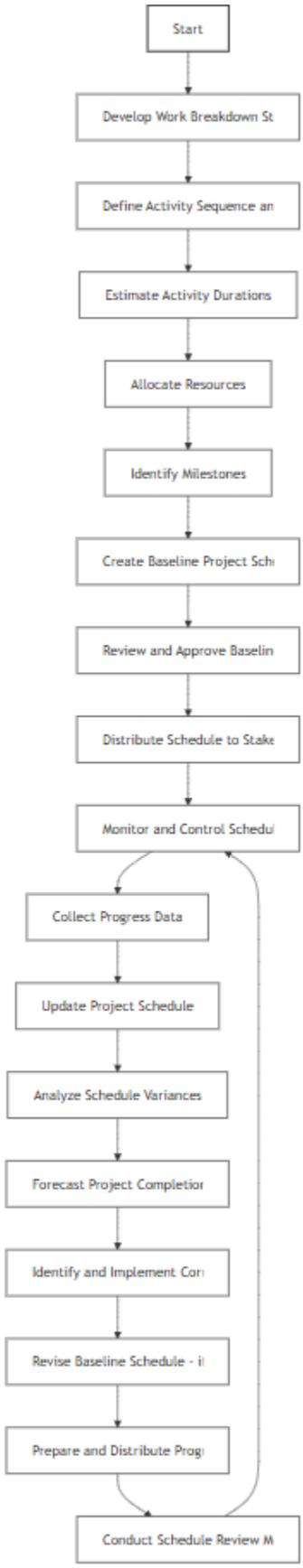
OBJECTIVE	This procedure outlines the process for developing, maintaining, and managing project schedules for BHBI construction projects. The aim is to ensure that projects are completed on time, within budget, and to the required quality standards by establishing a clear and realistic schedule that guides project execution.
SCOPE	This procedure applies to all construction projects undertaken by BHBI and covers all phases of the project lifecycle, from pre-construction planning through to project closeout. It encompasses the creation of baseline schedules, progress monitoring, schedule updates, and schedule analysis.

Responsibilities

- **Project Manager:** Overall responsibility for the development, implementation, and management of the project schedule.
- **Project Engineer:** Responsible for providing technical input to the schedule, monitoring progress against the schedule, and identifying any potential schedule deviations during the construction phase.
- **Site Engineer:** Responsible for providing on-site progress information to the Project Engineer for schedule updates.
- **Planning Engineer (or Scheduler):** If BHBI has a dedicated planning or scheduling role, this person is responsible for creating and maintaining the detailed project schedule using scheduling software.
- **Procurement:** Responsible for providing information on material and equipment delivery lead times to be incorporated into the project schedule.
- **Subcontractors:** Responsible for providing input on their activity durations and coordinating their schedules with the overall project schedule.



Procedure

4.1. Schedule Development:

- **Work Breakdown Structure (WBS):** The Project Manager, in collaboration with the Project Engineer and other key stakeholders, develops a comprehensive Work Breakdown Structure (WBS) that breaks down the project scope into manageable work packages or activities.
    - **Deliverables:** Work Breakdown Structure (WBS)
- **Activity Sequencing:** Define the logical sequence of activities, identifying dependencies between them (e.g., finish-to-start, start-to-start).

- **Duration Estimation:** Estimate the duration of each activity, considering factors such as resource availability, productivity rates, and potential risks. Involve relevant team members (e.g., Site Engineer, subcontractors) in duration estimation.
- **Resource Allocation:** Allocate resources (labor, equipment, materials) to each activity, considering resource availability and constraints.
- **Milestone Identification:** Identify key project milestones, such as contract award, start of construction, substantial completion, and project handover.
- **Baseline Schedule Development:** The Project Manager or Planning Engineer uses scheduling software (e.g., Microsoft Project, Primavera P6) to create the baseline project schedule, incorporating the WBS, activity sequences, durations, resource allocations, and milestones.
  - **Deliverables:** Baseline Project Schedule
- **Schedule Review and Approval:** The Project Manager reviews and approves the baseline schedule, ensuring that it is realistic, achievable, and aligned with the project objectives.

4.2. Schedule Implementation and Communication:

- **Schedule Distribution:** The Project Manager distributes the approved baseline schedule to all project stakeholders, including the client, consultants, subcontractors, and the project team.
- **Communication:** Clearly communicate schedule expectations and responsibilities to all team members.

4.3. Schedule Monitoring and Control:

- **Progress Data Collection:** The Site Engineer regularly collects data on actual progress achieved for each activity, including start and finish dates, percentage complete, and any delays encountered.
  - **Deliverables:** Progress Reports
- **Schedule Updates:** The Project Manager or Planning Engineer updates the project schedule regularly (e.g., weekly or bi-weekly) with actual progress data.
  - **Deliverables:** Updated Project Schedule
- **Progress Measurement:** Compare actual progress against planned progress to identify any deviations from the baseline schedule.
- **Variance Analysis:** Analyze schedule variances to determine the causes of delays or early completions and their potential impact on the project.
  - **Deliverables:** Schedule Variance Analysis
- **Schedule Forecasting:** Forecast the project completion date based on current progress and any identified variances.

4.4. Schedule Recovery and Corrective Actions:

- **Identify Corrective Actions:** If the project schedule falls behind, the Project Manager, in collaboration with the Project Engineer and relevant stakeholders, identifies and evaluates potential corrective actions. Corrective actions may include:
  - Re-sequencing activities
  - Adding resources
  - Working overtime
  - Adjusting activity durations
  - Fast-tracking (performing activities in parallel that were originally planned in sequence)
  - Crashing (adding resources to shorten activity durations)
- **Implement Corrective Actions:** The Project Manager implements the chosen corrective actions and monitors their effectiveness.
- **Schedule Revisions:** If necessary, the Project Manager revises the baseline schedule to reflect the impact of corrective actions or approved changes.

4.5. Schedule Reporting:

- **Regular Progress Reports:** The Project Manager prepares regular progress reports that include a summary of schedule performance, highlighting any variances, corrective actions taken, and a forecast of the project completion date.
  - **Deliverables:** Project Progress Reports
- **Schedule Review Meetings:** Hold regular schedule review meetings with the project team, client, and other stakeholders to discuss schedule status and address any issues.

4.6. Change Management:

- **Impact Assessment:** Any approved project changes (variations) that impact the schedule must be assessed for their impact on activity durations, dependencies, and the overall project completion date.
- **Schedule Updates:** The Project Manager updates the project schedule to reflect the impact of approved change orders.

4.7. Documentation and Record Keeping:

- **Schedule Records:** Maintain accurate records of all schedule-related information, including:
  - Baseline schedule
  - Schedule updates
  - Progress reports
  - Variance analyses
  - Corrective action plans
  - Schedule revisions
  - Change orders impacting the schedule

Continuous Improvement:

- **Lessons Learned:** Conduct post-project reviews to identify lessons learned regarding project scheduling.
- **Procedure Updates:** Periodically review and update this procedure to incorporate lessons learned and industry best practices.

Records Generated (Deliverables)

- Work Breakdown Structure (WBS)
- Baseline Project Schedule
- Progress Reports

- Updated Project Schedule
- Schedule Variance Analysis
- Project Progress Reports

Prepared by:	Reviewed by:	Approved by:	Noted by:
Marcone John Perea *Office Engineer*	**Mark Anthony Pajes** *Project Engineer*	**Nemesia P. Baylon** *Development Director*	**Adrian Carlo Escay** *President & CEO*